

VeXeRe: Helping millions of passengers annually take happy and safe bus trips with Google Cloud

Tell us your challenge. We're here to help.

Contact us (/contact)

ABOUT VEXERE



With stability, scalability, and advanced analytics from Google Cloud, VeXeRe enables millions of passengers to book bus tickets and purchase in just seconds, all while helping bus companies increase

About VeXeRe

Established in 2013, VeXeRe provides services to the inter-city bus industry in Vietnam through three solutions: a smart ticket booking platform for passengers, effective management software for bus operators (Bus Management System), and sales management software for agents (Agent Management System).

VeXeRe is the largest online bus ticketing platform in Vietnam with more than 700 bus companies, covering 2,600+ domestic and cross-border routes. VeXeRe Bus Management System

revenues by 30%.
more efficient op
enabled by digita
transformation.

(BMS) is used by more than 700 bus companies
while over 5,000 agents are using VeXeRe Agent
Management System to increase ticket sales
and serve their customers better.

Industries: Transportation

Location: Vietnam

Google Cloud results

- Provides the scalability to seamlessly support increases of up to 5x in demand for bus services during holidays
- Enables the business to use data to understand customer behavior and experiences and adapt

Improv
to 99.9
Google
meetir
custor
and ag
dema
reliabl
service

(<https://cloud.google.com/kubernetes-engine/>).



About Cloud Ace

Cloud Ace is a Google Cloud specialist cloud integrator that provides one-stop support, from cloud deployment planning to operation and maintenance. As a Managed Service Provider for Google Cloud, the business provides technical support, consulting, system development, and Google Cloud certified training. With offices in four cities in Japan and eight countries overseas, along with over 120 partner companies, the business aims to meet the diverse needs of companies undergoing digital transformation.



Google Cloud (<https://cloud.google.com/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

services
accordingly

- Ensures the business can quickly identify, navigate, and resolve any system issues through logging, monitoring, and tracing

Compute Engine

(<https://cloud.google.com/compute/>)

support from cloud deployment planning

Cloud Storage (<https://cloud.google.com/storage/>)

to operation and maintenance

BigQuery (<https://cloud.google.com/bigquery/>)

SQL data lake with integrated

Cloud SQL (<https://cloud.google.com/sql/>)

support, consulting, system development,

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Looker Studio

(<https://marketingplatform.google.com/about/data-studio/>)

meet the diverse needs of companies

undergoing digital transformation.

Founded in 2013, [VeXeRe](https://vexere.com/) (<https://vexere.com/>)

provides ecommerce and technology services to the inter-city bus industry in Vietnam, centered on three solutions: a smart ticket booking platform for passengers, management software for bus owners, and sales management software for agents.

The business provides the largest online bus ticketing and information platform in Vietnam, covering more than 700 bus companies and more than 2,600 domestic and cross-border bus routes. Passengers can select their seats and pay through a variety of methods, ranging from credit cards to e-wallets to cash, at convenience stores across the country.

Bus companies use the VeXeRe Bus Management System to modernize management, optimize revenue, and reduce costs, while 5,000+ agents use

the VeXeRe Agent Management System to increase ticket sales and serve customers better.

“At the time we were founded, there were few software solutions available to help bus companies, so we took the lead and now we have about 80% of the market in Vietnam,” explains Long Luong, Chief Technology Officer at VeXeRe.

VeXeRe initially ran its products and services in a hosted, bare-metal server infrastructure before moving to a private cloud in an internal data center. However, as traffic increased by 300%+ every year and the business added to the range and functionality of its products, demands on the infrastructure grew enormously. “We had difficulty scaling with our private cloud environment,” explains Long. “We have several holidays in Vietnam that see many people head from work in the cities to their hometowns, meaning demand for bus tickets increases three or five times normal levels.”

The business had to plan and prepare for these events by adding more server capacity one month ahead of time, enabling it to configure and test the stability of the updated infrastructure. Further, when the business experienced a sudden spike in traffic, it needed up to four hours to bring the required capacity online.

VeXeRe also experienced network and hardware issues in its private cloud environment that compromised availability and consequently customer experience, while its technology team also

had to restrict deployments to nights to avoid disrupting operations during the day.



“Given Google
originally
developed
Kubernetes—and
offered the
managed
Google
Kubernetes
Engine service—I
was confident
Google Cloud
was the best fit
for our
business.”

*—Long Luong, Chief
Technology Officer,
VeXeRe*

Access resources at scale

Long and his team explored the opportunities provided by multinational cloud services to access compute resources at scale and to automate the scaling of compute in line with demand. With the business running a microservices architecture to take advantage of ease of deployment, management, and looser dependencies, the Kubernetes (/kubernetes) open source container-orchestration system was an attractive, on-trend option.

“Given Google originally developed Kubernetes—and offered the managed Google Kubernetes Engine (/kubernetes-engine) service—I was confident Google Cloud was the best fit for our business,” says Long. “I also sought and received recommendations from peers to confirm my views and elected to proceed in early 2019.”

VeXeRe engineers with experience working with Google Kubernetes Engine—assisted by Google Cloud technical experts—helped complete the migration and the business elected to move its billing for the service to Google Cloud partner Cloud Ace (<https://www.cloud-ace.com/>).

“Cloud Ace provided extensive support, including cost breakdowns and monitoring and making recommendations about best practices, such as making use of committed-use discounts and optimizing autoscaling,” says Long. “We continue to stay in touch with the Cloud Ace team—as well as their expertise with Google Cloud—we learn from events they conduct that gather the local technology community together to listen to case studies on best practices.”

“With Google
Cloud, we can
start the
deployment
processes
associated with
launching a new
product or even
a new business
without creating
system
infrastructure
ahead of time as
we had to
previously.”

*—Long Luong, Chief
Technology Officer,
VeXeRe*

Increased uptime to 99.98%

Thanks to Google Kubernetes Engine and Google Cloud services overall, VeXeRe has increased uptime from 99.9% or less in the bare metal and private cloud environments, to 99.98% with Google Cloud. “The network and hardware issues we experienced in our private cloud have disappeared with Google Cloud,” explains Long.

Furthermore, Google Cloud enables the business to automate the deployment of new services when stable and shutting down old services, improving efficiency and eliminating the risk of disruption during these processes.

In addition, autoscaling ensures VeXeRe can effectively align cost with resource usage—a capability that proved particularly important when the government shut down the nation’s transport systems to combat spread of the coronavirus throughout the country. Long acknowledges this event would have proven expensive for the business had its private cloud still been operating. “We couldn’t minimize our costs as needed when traffic dropped during this period, because we had fixed six- to 12-month rental agreements for our servers,” says Long.

VeXeRe also relies on Google Cloud's operations suite for the logging, monitoring, and tracing needed to ensure system stability. "We know that, with Google Cloud, if one day something happens, we can quickly identify and navigate to the issue so we can resolve it," says Long.

Compute Engine (/compute) provides the required compute resources while Cloud Storage (/storage) removes impediments and limitations the business experienced while running a home-grown solution. "Maintaining our previous storage solution took time and lacked the fine-grained permission controls we needed to ensure teams could access only the folders or files they needed to," says Long. "Cloud Storage enabled us to resolve these issues and, together with the support of file events that connected easily to Pub/Sub (/pubsub), enabled us to quickly bootstrap a data pipeline to parse CSV - Microsoft Excel files and put them into a BigQuery (/bigquery) data warehouse for deeper analysis."

"We are
constantly
looking for new
business
opportunities,
and Google
Cloud enables

us to focus on
improving our
products and
developing new
products for new
customer
segments. And
we can do this
without worrying
about scaling to
support more
users.”

*—Long Luong, Chief
Technology Officer,
VeXeRe*

Creating new lines of business

Google Cloud also gives VeXeRe the agility to seamlessly create new lines of business, such as expanding its platform to offer goods transport, to seize market opportunities and dilute the impact of the pandemic in its operations. “With Google Cloud, we can start the deployment processes associated with launching a new business without creating system infrastructure ahead of time as we had to previously,” explains Long. “We are constantly looking for new business opportunities, and Google Cloud enables us to focus on improving our products and developing new products for new

customer segments. And we can do this without worrying about scaling to support more users.”

With agility, scalability, and stability meeting VeXeRe’s requirements, the business is eyeing Google Cloud’s data analytics ecosystem as a key source of competitive advantage. “We live in a time of data—every company generates data, but the company that makes best use of data wins the market,” explains Long. “With Google Cloud, we have been building a data platform to help us understand customer segmentation and behavior, so we can provide the best recommendations when they browse our service.”

The business uses Analytics 4—the next generation of Google Analytics, that uses machine learning to deliver insights and understand customer behavior across devices and platforms—to export raw events into BigQuery to be queried, while monitoring allows it to track the quality of the customer experience, including the speed of customer interactions with the VeXeRe platform, and make any required adjustments.

Insights derived from BigQuery, such as the percentage of customers experiencing low response times for products and the individual customers that needed to be contacted for further investigation, are visualized through Looker Studio (<https://marketingplatform.google.com/about/data-studio/>)

for ease of understanding and fast, accurate decision-making by business leaders. VeXeRe has moved most of its data to a relational database in

Cloud SQL (/sql) to use the efficiency and scalability of the service, while BigQuery-Cloud SQL federation enables the business to use the data analytics warehouse to query data residing in the database in real time, without any copying or movement.

VeXeRe is now looking at the opportunities Google Cloud's AI and machine learning products provide to further enhance its capabilities. "With Google Cloud, we're ideally positioned to broaden our offerings, provide even better customer experiences, and contribute to community and business life in Vietnam," says Long.

